Case Study: NLP-Based Public Grievance Classification and Priority Detection

1. Problem Statement

Public grievance redressal systems across the world, and in India in particular, face a structural mismatch between the volume and complexity of citizen complaints on one side and the capacity of manual, human-driven triage processes on the other. Platforms such as the Centralised Public Grievance Redress and Monitoring System (CPGRAMS) connect citizens to essentially every ministry, department, and state government body in the country through a single portal, and have processed over 11.2 million grievances between 2019 and 2024 alone. At this scale, sorting complaints by hand into the correct department, correctly identifying which complaints are genuinely urgent, and doing all of this across dozens of languages and dialects, including code-mixed forms such as Hindi-English Hinglish, is no longer operationally feasible using manual, rule-based methods.

The problem is not simply one of volume. It is compounded by three additional factors that manual and purely rule-based systems handle poorly. First, linguistic diversity: India alone recognises 22 scheduled languages, and citizens frequently write complaints in regional languages, indigenous languages such as Bhojpuri, Garo, and Khasi, or in code-mixed text that blends Hindi and English within the same sentence. Second, category imbalance: in real-world complaint datasets, a small number of categories account for the overwhelming majority of complaints while many other categories, some of them safety-critical, appear rarely. This has been documented empirically, for example in a 2026 study of Indonesia's LaporGub Central Java e-governance platform, which found an imbalance ratio of approximately 227 to 1 across 18 complaint categories in a dataset of 53,877 real complaints. Third, urgency detection: not all grievances carry equal weight, and a system that cannot reliably distinguish a life-threatening or rights-violating complaint from a routine service request risks delaying the cases that matter most, a failure mode that rule-based keyword matching alone struggles to avoid.

The core problem that motivates NLP-based grievance classification and priority detection, and that the NyayaGrievance project is built to address, can therefore be stated directly: manual, rule-based grievance triage does not scale to multilingual, code-mixed, high-volume citizen input, and without automated classification and urgency detection, government grievance systems risk slower resolution times, misrouted complaints, and delayed attention to genuinely urgent cases.

2. Theory

The theoretical grounding for NLP-based grievance classification rests on three related but distinct bodies of academic work: multilingual and code-mixed text classification, priority or urgency detection, and, more recently, large language model and zero-shot approaches that reduce the need for bespoke, per-department classifiers.

2.1 Multilingual and Code-Mixed Classification

A foundational difficulty in grievance classification is that conventional, single-language machine learning models struggle to capture the semantic nuance of code-mixed text, where a complaint may switch between two or more languages within the same sentence or even the same phrase. A 2024 arXiv paper on Hinglish cybercrime complaint classification frames this problem precisely, arguing that manual triage of large complaint volumes is inefficient and that standard classifiers are not equipped to handle the code-switching common in Hindi-English text. This motivated the development of Hinglish-adapted transformer models, HingBERT and HingRoBERTa, which were later produced as real-world datasets and models through the Indian Cybercrime Coordination Centre's CyberGuard AI Hackathon in 2024.

A separate but equally important theoretical challenge is extreme class imbalance. The 2026 study of the LaporGub platform in Central Java compared a traditional TF-IDF and Support Vector Machine baseline against XLM-RoBERTa, both with and without class-weighted loss functions. The finding that raw model architecture matters less than correctly handling skewed real-world category distributions is an important theoretical lesson: a technically sophisticated transformer model can still perform poorly on rare-but-important categories if the imbalance in the training data is not explicitly corrected for.

2.2 Priority and Urgency Detection

A second pillar of the theory concerns how a system distinguishes urgent complaints from routine ones. A 2026 IJERT paper on a Civic Complaint Priority Management System combines three complementary techniques: rule-based keyword matching implemented through RapidFuzz, a TF-IDF plus Logistic Regression fallback model, and a location-density signal that accounts for where complaints are geographically clustered. This hybrid approach reported 91.3% overall accuracy and, more importantly for urgency detection specifically, 94.2% recall on the highest-priority complaint class, outperforming both a pure rule-based system and a pure machine learning baseline.

This theoretical direction, that hybrid or transformer-based urgency detection outperforms naive keyword rules, is reinforced outside the Indian and English-language context as well. A ScienceDirect study applied BETO, a Spanish-language BERT model, to binarily classify telecom-style complaints as urgent or not, achieving 92.1% accuracy and an area under the curve of 0.9785. This is theoretically significant because it demonstrates that transformer-based urgency detection generalises across languages rather than being an artefact specific to English or Hindi text.

A 2025 Nature Scientific Reports paper extends the theory further by proposing a zero-shot framework that combines MobileBERT for semantic routing, a CTC-trained recurrent neural network to transcribe multilingual and dialect-rich voice complaints, and lexicon-based sentiment analysis feeding into time-aware escalation logic. This points toward voice-based, dialect-tolerant grievance intake as a plausible theoretical next stage beyond purely text-based classification.

2.3 Large Language Model and Zero-Shot Approaches

The most recent theoretical development in this space is the shift from training bespoke classifiers per department or per category toward using large language models and instruction-tuned embedding models directly. A 2025 MDPI study compares large language models against instruction-tuned embedding models for categorising public transport complaints, motivated by the observation that public agencies need classification that is both accurate and operationally efficient. This same accuracy-versus-efficiency trade-off underlies the design choice, discussed further in the description section below, to use a hosted large language model rather than training and maintaining a separate classifier for every government department.

This body of literature builds on earlier foundational work. A 2022 study proposing an NLP-based model for classifying 10,000 education-department complaints established the baseline theoretical claim that underlies all of the more recent research: even simple supervised classifiers substantially outperform manual sorting. Every transformer-based and large language model-based approach discussed above is, in theoretical terms, an improvement upon that original baseline rather than a departure from it.

3. Description

This section describes the government systems and academic studies that constitute the evidence base for NLP-based grievance classification, and then describes how the NyayaGrievance project fits within this landscape.

3.1 CPGRAMS, India's Central Grievance Portal

The Centralised Public Grievance Redress and Monitoring System, run by the Department of Administrative Reforms and Public Grievances, connects citizens to essentially every ministry, department, and state government body in India through a single 24x7 portal, also reachable via the UMANG mobile application. Government reform documentation describes a deliberate move toward artificial intelligence within this system: automatic flagging of urgent grievances using AI and machine learning, intelligent monitoring dashboards, spam identification, and improved auto-routing of complaints to the correct last-mile officer, alongside portal support in 22 scheduled languages plus English.

CPGRAMS is described as one of the largest grievance redress initiatives globally, having processed over 11.2 million grievances between 2019 and 2024, with reported reductions in average resolution time as these reforms were rolled out. In 2026, CPGRAMS introduced an AI-enabled chatbot named Samadhan Didi, which combines Bhashini's language capabilities with grievance-classification models trained on CPGRAMS's own complaint data, extending language support beyond the 22 scheduled languages toward regional and indigenous languages such as Bhojpuri, Garo, and Khasi.

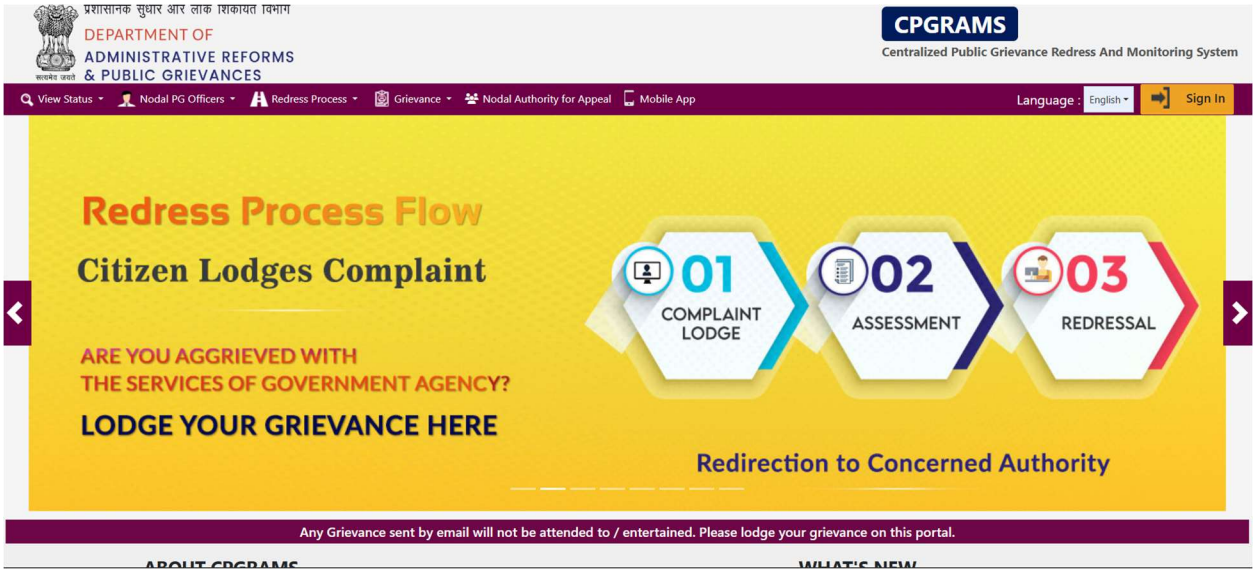


Fig 3.1.1 CPGRAMS portal

3.2 Bhashini, the National Language Translation Mission

Bhashini, which stands for Bhasha Interface for India, was launched in 2022 under the Ministry of Electronics and Information Technology and serves as the language-technology backbone that other government platforms, including the newer CPGRAMS chatbot, plug into. It is positioned as public API infrastructure rather than a single standalone application, covering translation, speech recognition, and text-to-speech across the 22 constitutionally recognised languages. Bhashini also runs a crowdsourcing arm called BhashaDaan, used to gather voice, text, and image data for under-resourced languages such as Konkani and Bodo.

This is a useful description for any grievance system claiming multilingual capability, because Bhashini demonstrates that the Indian government itself treats large-scale multilingual natural language processing as shared infrastructure to be integrated, rather than something to be rebuilt from scratch by each individual department. This is the same architectural choice that the NyayaGrievance project makes, by relying on a large language model's native multilingual capability rather than constructing a bespoke translation layer.

3.3 Sector-Specific Deployments

- Food Corporation of India Grievance Redressal System (2024): a domain-specific grievance channel built for food procurement, aimed at rice millers and grain distribution stakeholders, showing that the CPGRAMS model is being replicated in specialised sectors rather than treated as a one-size-fits-all solution.
- Indian Cybercrime Coordination Centre CyberGuard AI Hackathon (2024): produced real-world datasets and models, HingBERT and HingRoBERTa, specifically for classifying Hindi-English code-mixed cybercrime complaints, direct evidence that code-mixed complaint classification is an active government-recognised problem rather than a hypothetical one.

- National Consumer Helpline: reported to use AI-based text analytics to bucket complaints into themes such as service delays, payment failures, or fraud, alongside CPGRAMS.

3.4 International Comparator: NYC 311

Outside India, New York City's 311 non-emergency service request system is the most heavily studied grievance-style platform in the academic literature. It receives more than eight million requests a year and has been used as a testbed for gradient boosting models predicting building code violations, neural networks classifying which city agency should handle a complaint, reaching approximately 90% test accuracy in one published study, and equity research examining whether complaint volume under- or over-represents actual problems across different neighborhoods.

The equity dimension of this research is a useful caution for any classifier-based grievance system. Research on NYC 311 explicitly warns that models trained on historical complaint data can inherit reporting bias, since the propensity to file a complaint is not evenly distributed across a city's population.

3.5 Comparative Summary of Systems and Studies

Table 1 below brings together the government systems and academic studies described above, along with their core NLP approach, reported results, and language scope, to allow direct comparison across deployments.

System / Study	Core NLP Approach	Reported Result	Language Scope
CPGRAMS (DARPG, Government of India)	AI/ML auto-flagging of urgent grievances; auto-routing to last-mile officers	Reforms cut average resolution time; over 11 million grievances processed 2019 to 2024	22 scheduled languages plus English
CPGRAMS 'Samadhan Didi' chatbot (2026)	Bhashini language layer combined with grievance-classification models trained on CPGRAMS data	Phased rollout adding regional and indigenous languages	22+ languages, expanding
I4C CyberGuard AI Hackathon (HingBERT / HingRoBERTa)	Transformer models adapted for Hinglish code-mixed text	Improved crime-type classification on real complaint data	Hindi-English code-mixed
LaporGub Central Java e-governance platform study	XLM-RoBERTa with class-weighted cross-entropy compared against TF-IDF plus SVM	Better fairness on a 227:1 imbalanced, 18-category dataset of 53,877 complaints	Indonesian
Civic Complaint Priority Management System (IJERT, 2026)	Hybrid rule-based matching plus TF-IDF/Logistic Regression plus location density score	91.3% overall accuracy; 94.2% recall on critical (P1) complaints	English

Analytics-driven complaint prioritisation (ScienceDirect)	BETO, a Spanish-language BERT model, used as a binary urgency classifier	92.1% accuracy, 0.9785 AUC	Spanish
Zero-shot multimodal grievance framework (Nature Scientific Reports, 2025)	MobileBERT zero-shot classification, CTC-based speech transcription, and lexicon-based sentiment scoring for urgency	Unified pipeline for typed and spoken, multilingual and dialect input	Multilingual, voice-inclusive
NYC 311 (multiple academic studies)	Gradient boosting and neural network models for agency routing and equity analysis	Approximately 90% agency-routing accuracy in one study; used for resource allocation	English

Table 3.5.1. Selected government systems and academic studies on grievance and complaint NLP, and how their approach and results compare.

3.6 Where NyayaGrievance Sits in This Landscape

Read against this evidence base, the architectural choices made in the NyayaGrievance project track the direction the field and government deployments are already moving in, rather than reinventing them from first principles.

- Multilingual handling via a large language model's native capability, specifically LLaMA models hosted through Groq, mirrors the large language model and embedding comparison literature and the government's own move, seen in Bhashini being plugged into the CPGRAMS chatbot, toward treating multilingual understanding as a capability of the underlying language model or shared infrastructure rather than a hand-built per-language pipeline. This is an important and honest technical distinction to keep in any write-up of the project: it is large language model-native multilingual competence, not an engineered translation layer built specifically for the project.
- Department classification framed as multi-class text classification is the same framing used in the CPGRAMS auto-routing reforms, the LaporGub study, and the education-complaint classification paper, making it a well-validated approach rather than a novel one.
- Priority detection via linguistic urgency cues sits in the same family as the IJERT hybrid system and the BETO urgency classifier, both of which show that hybrid or transformer-based urgency detection reliably outperforms naive keyword rules.
- The WhatsApp and Twilio-based communication channel is a genuine point of difference from CPGRAMS, which relies on a web portal and the UMANG app, and from NYC 311, which relies on call, text, and web channels. This lowers the barrier for citizens who are already comfortable with WhatsApp, and is similar in spirit to the voice-based, low-friction intake that the 2025 Nature paper argues for.

4. Conclusion

The case for NLP-based grievance classification and priority detection does not rest on a single research paper. It is corroborated independently by a national government's own reform documentation, seen in CPGRAMS and Bhashini, by a purpose-built hackathon dataset created specifically for code-mixed complaints through the I4C CyberGuard initiative, by multiple peer-reviewed and preprint studies published between 2025 and 2026 across several languages including Hindi-English, Indonesian, and Spanish, and by roughly a decade of NYC 311 research conducted in a non-Indian context. Read together, this evidence supports the same conclusion that the NyayaGrievance project is built on: manual, rule-based grievance triage does not scale to multilingual, code-mixed, high-volume citizen input, and hybrid or large language model-based NLP pipelines consistently outperform manual triage on both routing accuracy and urgent-case recall.

At the same time, an honest reading of this literature requires naming several open research gaps rather than overstating what has been proven. Reporting bias remains a real risk, since NYC 311 equity research shows that complaint volume does not evenly reflect underlying problems across a population, a risk that applies equally to CPGRAMS-scale and NyayaGrievance-scale deployments if raw complaint counts are treated as ground truth for resource allocation. Class imbalance is another persistent issue, illustrated starkly by the LaporGub study's 227 to 1 ratio, which is a reminder that department categories in most real deployments are far from evenly distributed and can silently degrade accuracy on rare-but-important categories unless this imbalance is explicitly corrected for during training. Code-mixing and dialectal variation remain an open technical challenge as well, since Hinglish-specific transformer models exist precisely because general-purpose multilingual models still underperform on code-mixed and dialectal input, a gap that the government's own phased BhashaDaan data-collection effort is trying to close for languages such as Konkani and Bodo.

Finally, evaluation transparency deserves explicit mention. Several of the more polished accuracy figures cited throughout this case study, including 91.3%, 92.1%, and approximately 90%, come from held-out test sets containing a few hundred to a few thousand labelled complaints. None of the papers reviewed here report large-scale, multi-year field validation at the scale of CPGRAMS itself, which remains an open question for any academic claim of real-world readiness. Naming these gaps honestly does not weaken the case for NLP-based grievance classification and priority detection; it strengthens it, by distinguishing what is genuinely well-established in the literature and in government practice from what remains to be proven at scale, and by situating the NyayaGrievance project as a system that applies well-validated techniques through a genuinely differentiated communication channel, rather than one that claims to have solved problems the wider field has not yet solved.